

Assignment - 01
Computer Network

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• Sec - A

(22BCON1068)

(1) What is Protocol?

Ans:- Protocol is a set of rules that governs the communications between computers on a network. These rules include guidelines that regulate the following characteristics of a network: access method, allowed physical topologies, speed of data transfer.

Q 2. What are the service provided by data link layer?

Ans:- The primary service of the data link layer is to support error-free transmission. The physical layer sends the data from the sender's hub to the receiver's hub as raw bits. The data link layer should recognize and correct some error in the communicated data.

Q 3. What is data communication? Write components of data communication.

Ans:- Data communication is the process of sending or receiving digital data between two or more computers via transmission medium such as a wire cable or wireless.



• Component of data Comm. :-

Data Communication system has five components.

- (i) Message (ii) Sender (iii) Receiver
- (iv) Transmission medium
- (v) Protocols.

Q4. Write the difference betⁿ. Asynchronous and Synchronous transmission.

Ans. The diff. betⁿ. asynchronous and Synchronous transmission is given below -

Asynchronous Transmission	Synchronous Transmission
(i) Transmits data in the form of frames or blocks.	(i) Transmits 1 byte
(ii) Transmits 1 byte or character at a time.	(ii) Transmits data in the form of frames or blocks.
(iii) Transmission speed is slower.	(iii) Transmission speed is faster.
(iv) Cost is economical.	(iv) Cost is expensive.
(v) Time interval is random.	(v) Time interval is constant.
(vi) There are no gaps bet ⁿ . the data.	(vi) There are gaps bet ⁿ . the data.
(vii) Used in Email, Forums, Letter, etc.	(vii) Used in chat Rooms, Telephone conversation, video conferencing.

Q5. What is Internet ?

Ans. The internet is a vast network that connects computers all over the world. Through the internet, people can share information and communicate from anywhere with an internet connection.

In other words -

The Internet is defined as a global network of linked computers, servers, phones, and smart appliances that communicate with each other using the TCP standard to enable the fast exchange of information and files, along with other types of services.

Sec - B : (7x1)

Q1. What is Computer Network? Also describe the types of Network.

Ans. A computer network is a group of computers linked to each other that enables the computer to communicate with another computer and share their resources, data and applications.

A computer network is mainly five types.

(i) PAN (Personal Area Network):- PAN is the most basic type of computer network. This network is restrained to a single person, that is communication betⁿ computer devices is centred only on an individual's workspace. PAN offers a network range of 1 to 100 meters. Like Bluetooth, etc.

(ii) LAN (Local Area Network):- A LAN is a computer network that connects computers through a common communication path, contained within a limited area, that is locally. A LAN encompasses two or more computers connected over a server. The two important technologies involved in this network are "Ethernet" and "Wi-Fi". It ranges up to 2 km & transmission speed is very high with easy maintenance and low cost.

(iii) CAN (Campus Area Network):- This is a type of computer network that is usually used in places like a school or colleges. This network covers a limited geographical area that

is, it spreads across several building within the campus. CAN mainly use Ethernet technology with a range from 1 km to 5 km.

Its transmission speed is very high with a moderate maintenance cost and moderate cost.

(iv) Metropolitan Area Network (MAN):- This is a type of computer network that connects computers over a geographical distance through a shared communication path over a city, town or metropolitan area. This network mainly uses FDDI, CDDI, and ATM as the technology with a range from 5 km to 50 km. Its transmission speed is average. It is difficult to maintain and it comes with a high cost.

(v) WAN (Wide Area Network):- WAN is a type of computer network that connects computers over a large geographical distance through a shared communication path. It is not restrained to a single location but extends over many locations.

A Wide Area Network is a network that extends over a large geographical area such as states or countries. The Internet is one of the biggest WAN in the world. A Wide Area Network is widely used in the field of Business, Government, and Education. Examples of WANs are mobile Broadband, last mile, etc.

Q 2:

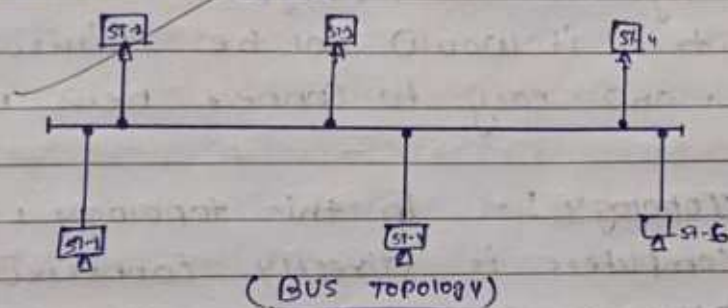
What is Topology? Explain all the types of topology with diagram.

Ans:

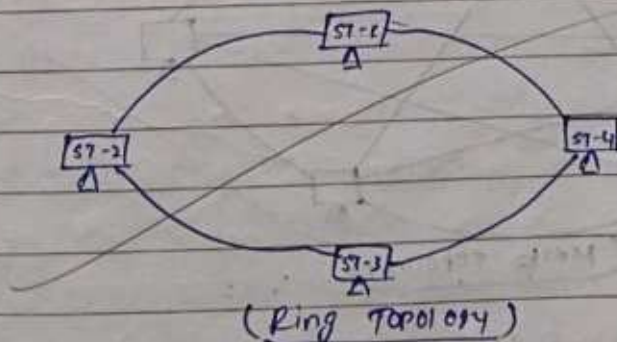
The Physical Arrangement of Computer System/node which is connected to each other via Communication medium is called Topology.

The Physical topology mainly four types that is following -

- (i) BUS Topology:- in Bus topology, one long cable acts as a single communication channel and all the devices are connected to this cable.

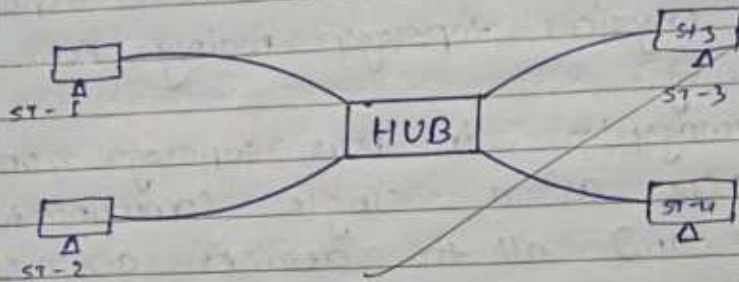


- (ii) Ring Topology:- it is called Ring topology because it forms a ring. in this topology each node is strongly connected with its adjacent node.



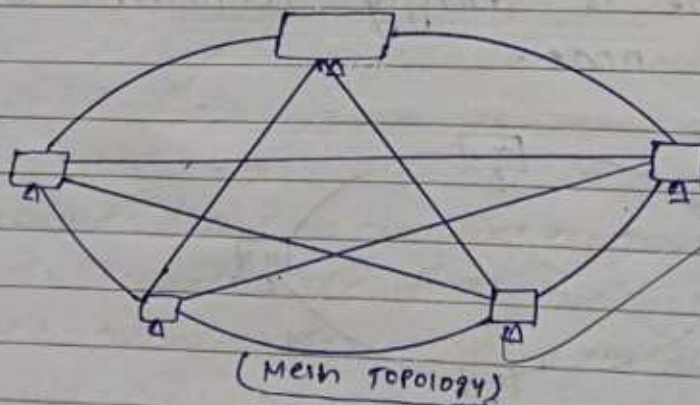
The advantages of Ring topology is Transmission rate of data is very speed and disadvantage is, it is very difficult task to add some new computers.

- (iii) STAR Topology :- in star topology all the nodes are connected with a central device HUB. and the sharing the data is only possible through HUB.



advantage of STAR Topology is that, if one node fails, then it would not be failure of entire network. and easy to connect new nodes.

- (iv) MESH Topology :- In this topology, each and every computer is directly connected with each other. So, we can directly send the data to the destination machine without going to intermediate machine.



Advantage :- if multiple devices can send or receive data simultaneously.
 → if a particular machine is failed then entire network will not fail.

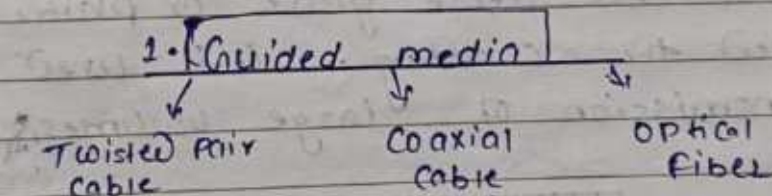
Q3. What is transmission media? Explain its type in detail.

Ans. In data communication terminology, a transmission medium is a physical path betⁿ the transmitter and the receiver. That means it is the channel through which data is sent from one place to another.

Transmission media is broadly classified into two types.

(1) Guided media : (2) Unguided media.

(1) Guided media :- it is also referred to as wired or Bounded transmission media. Signals being transmitted are directed and confined in a narrow pathway by using physical links. There are 3 major types of Guided media -



(i) Twisted Pair Cable :- it consists of 2 separately insulated conductor wires wound about each other. Several such pairs are bundled together in a protective sheath.

Twisted pair is of two types :-

(a) UTP (Unshielded Twisted Pair) :- UTP consists of two insulated copper wires around one another.

(b) Shielded Twisted Pair (STP) :- This type of cable consists of a special jacket (a copper braid or a foil shield) to block external interference.

- (ii) CO-axial Cable :- it has an outer plastic covering containing an insulation layer made of PVC or Teflon and 2 parallel conductors each having a separate insulated protection cover. The CO-axial cable transmits information in two modes: Baseband mode and Broadband mode. Cable TVs and analog television networks widely use co-axial cables.

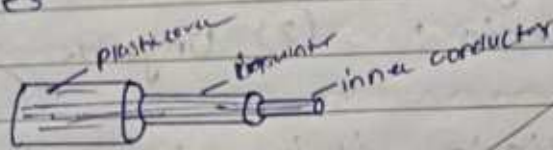
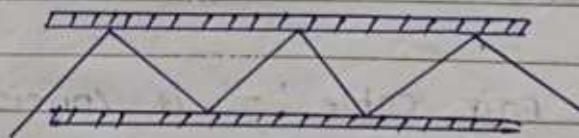


Figure 1. CO-axial Cable

- (iii) Optical Fiber Cable :- it uses the concept of refraction of light through a core made up of glass or plastic. The core is surrounded by a less dense glass or plastic covering called the cladding. It is used for the transmission of large volumes of data.

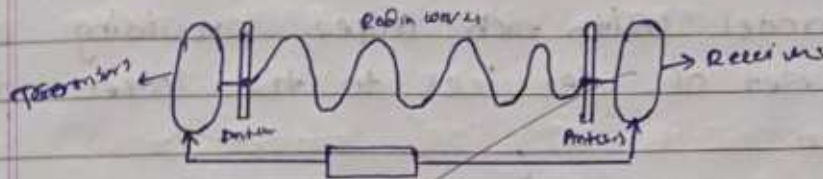


2. Unguided media :- These are easy to generate and can penetrate through buildings. The sending and receiving antennas need not be aligned.

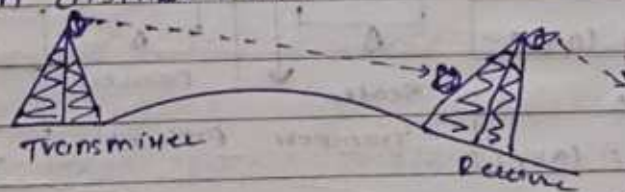
→ it is also referred to as wireless or unbounded transmission media. NO physical medium is required for the transmission of electromagnetic signals.

There are 3 types of unguided media :-

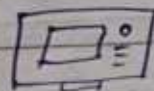
- (i) Radio Waves :- These are easy to generate and can penetrate through buildings. The sending and receiving antennas need not be aligned. Frequency range : 3 KHz - 1 GHz. AM and FM radio and Cordless Phones use radio waves for transmission.



- (ii) Microwaves :- The sending and receiving antennas need to be properly aligned with each other. The distance covered by the signal is directly proportional to the height of the antenna. Frequency Range : 1 GHz - 300 GHz. These are majority used for mobile phone comm. and Television distribution.



- (iii) Infrared :- infrared waves are used for very short distance communication. This prevents interference betⁿ system. Frequency Range: 300 GHz - 400 THz. It is used in TV remotes, wireless mouse, keyboard, Printer.



television



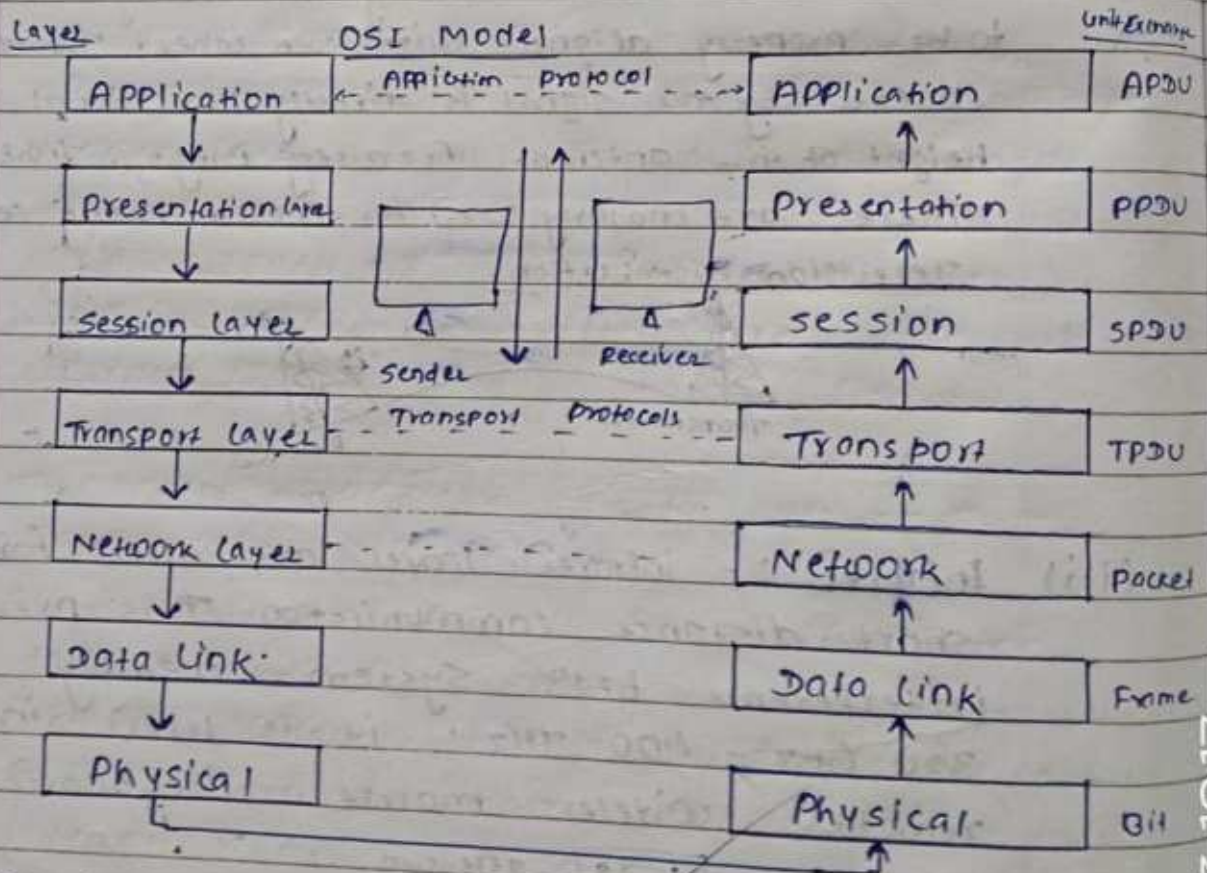
infrared Radiations.

Section - C (3x11)

Q (1). What is OSI Model. Explain the functions and protocols and service of each layer? Compare with TCP/IP Model.

Ans. The Open System Interconnection (OSI) model is a conceptual model that describes how data is communicated over a network. It is a layered model with each layer providing a specific set of services to the layer above it.

The OSI model has seven (7) layers.



(i) Physical Layer:- The physical layer is responsible for transmitting raw bits over a physical medium, such as a copper cable or fiber optic cable.

- (ii) Data Link Layer :- The data link layer provides reliable transmission of data over the physical layer. it does this by using error detection and correction mechanisms.
- (iii) Network Layer :- The network layer is responsible for routing data packets from the source to the destination. it does this by using a routing table to determine the best path for the data to take.
- (iv) Transport Layer :- The transport layer is responsible for routing and provides reliable end-to-end communication betⁿ. applications. it does this by using flow control and error recovery mechanisms.
- (v) Session Layer :- The session layer establishes, manages, and terminates communication sessions between applications. it also provides synchronization and recovery mechanisms.
- (vi) Presentation Layer :- The presentation layer translates data betⁿ. different formats so that it can be understood by the receiving application.
- (vii) Application Layer :- The application layer provides services to end-user applications, such as file transfer, email, and web browsing.

• Protocols and Services of Each Layer :-

Each layer of the OSI model uses a set of protocols to provide its services. some common protocols used in the OSI model include -

- Physical Layer:- Ethernet, Token Ring, Fiber Optic
- Data Link Layer:- HDLC, PPP
- Network Layer:- IP, ICMP, IGMP
- Transport Layer:- TCP, UDP
- Session Layer:- NetBIOS, RPC
- Presentation Layer:- ASCII, JPEG, MPEG
- Application Layer:- HTTP, FTP, SMTP, DNS

Comparison with TCP/IP model:-

OSI model		TCP/IP Model
Application Layer		Application layer
Presentation Layer	←→	
Session Layer		
Transport Layer	←→	Transport layer
Network Layer	←→	Internet layer
Data Link Layer	←→	Network access
Physical Layer		Layer

The TCP/IP model is a simplified version of the OSI model. It has 4 layers.

- Application layer:- similar to the application layer of the OSI model.
- Transport layer:- similar to the transport layer of the OSI.
- Internet layer:- similar to the network layer of the OSI.
- Network access layer:- similar to the data link and physical layer of the OSI model.

Differences:-

The main difference betⁿ the OSI and TCP/IP model is that the OSI model is more detailed and comprehensive. It provides a more complete description of how data is communicated over a network. The TCP/IP model is more simplified and focused on the specific protocols that are used on the internet.

OSI model	TCP/IP model
(i) it is developed by ISO. (International standard organization)	it is developed by ARPANET.
(ii) OSI refers to Open System interconnection.	TCP refers to transmission control protocol.
(iii) It follows vertical approach.	It follows a horizontal approach.
(iv) Session and Presentation layer are a part of the OSI model.	There are no session and presentation layer in the TCP/IP layer.
(v) it has seven layers.	it has four layers.

Q.2. Let us assume the even parity hamming code from the above example.

(111001101) is transmitted and the received code is (110001101). Now from the received code, let us detect and correct the error.

Solⁿ

(i) Received code :- (110001101)

(ii) Verify Even parity :-

Count the number of ones in the received

Code, including the parity bit.

In even parity,

this count should be even. If it's odd, it indicates that an error has occurred.

- Received code :- (110001101)
- Number of ones: ~~6 (even)~~ 5 (odd).

Since the no. of ones is even, this suggests that the parity check is satisfied. Which means there might not be an error.

If no. of ones is odd, this means there is an error.

- For Received code (110001101), the Parity bits are

- $P_1 = 1(1+0+1+1=3)$
- $P_2 = 1(1+0+1+1=3)$
- $P_3 = 1(0+1+1+1=3)$

- (iii) Compare the calculated parity bits to the received parity bits.

In this case,

received Parity bits are $P_1 = 1, P_2 = 0, P_3 = 1$

therefore, there is an error in the received code.

- (iv) Locate the error in the received code.

- To locate the error in the received code, we can use the following table -

Error syndrome	Bit error
000	No error
001	Bit 1
010	Bit 2
011	Bit 3
100	Bit 4
101	Bit 5
110	Bit 6
111	Bit 7

The error syndrome for the received code is 010, which corresponds to bit 2 in error.

(V) Correct the error in the received code :-

To correct the error in the receive code we simply flip the bit in error. In this case we flip bit 2 from 0 to 1.

The corrected received code is therefore (111001101).

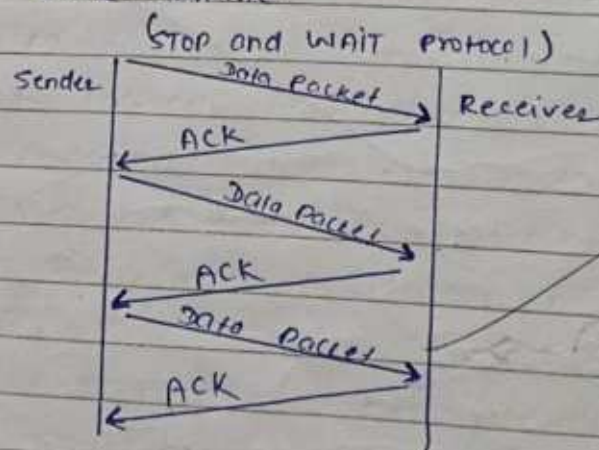
AS

(PTO) →

Q3. Write a short notes on :-

(i) STOP & WAIT Protocol :- Stop and wait Protocol is a simple automatic Repeat Request (ARQ) Protocol used for reliable data transmission over an unreliable channel. (It is a simplex Protocol, meaning the data can only be transmitted in one direction at a time.) In stop and wait, the sender transmits a single frame and then waits for an acknowledgement (ACK) from the receiver. If the sender does not receive an ACK within a certain timeout period, it re-transmits the frame. Once the receiver receives the frame correctly it sends an ACK back to the sender. The sender then transmits the next frame in the sequence.

STOP and wait is a simple protocol, but it is also inefficient. This is because the sender can only transmit one frame at a time, and it has to wait for an ACK before transmitting the next frame.



The above figure shows the working of the stop and wait Protocol.

• Advantages of STOP and wait Protocol: -

- simple to implement
- low overhead

• Disadvantages: -

- problems occur due to lost data and acknowledgement.
- Not suitable for unreliable channels.

∴ (ii) Sliding Window Protocol :- The sliding window is a technique for sending multiple frames at a time. It controls the data packets between the two devices where reliable and gradual delivery of data frames is needed. It is also used in TCP (Transmission Control Protocol).

In this technique, each frame has sent from the sequence number. The sequence numbers are used to find the 'missing data' in the receiver end. The purpose of the sliding window technique is to avoid duplicate data, so it uses the sequence number.

Sliding window protocols work by maintaining a window of frames that the sender can transmit without for an ACK. The size of the window is determined by the bandwidth of the channel and the delay involved in transmitting data over the channel.

- Advantage:
 - more efficient than stop & wait.
 - suitable for unreliable channel
 - can achieve higher throughput.

- Disadvantages:
 - more complex to implement than stop & wait.
 - Higher overhead due to the need to maintain a window of frames.

- Types of sliding window protocol:
 - ↳ Sliding window protocol has two types:
 1. Go-Back-N ARQ
 2. Selective Repeat ARQ

(1) Go-Back-N-ARQ :- Go-Back-N-ARQ Protocol is also known as Go-Back-N-Automatic Repeat Request. It is a data link layer protocol that uses a sliding window method. In this, if any frame is corrupted or lost, all subsequent frames have to be sent again. The size of the sender window is N in this protocol. For ex - 8, the size of the sender window will be 8. The receiver window size is always 1.

(2) Selective Repeat ARQ :- It is a data link layer protocol that uses a sliding window method. The Go-Back-N-ARQ Protocol works well if it has fewer errors. But if there is a lot of error in the frame, lots of bandwidth loss in sending the frames again. So, we use the Selective Repeat ARQ Protocol. In this protocol, the size of sender window is always equal to the size of the receiver window. The size of the sliding window is always greater than 1.

If the receiver receives a corrupt frame,

it does not directly discard it. it sends a negative acknowledgement to the sender.

- Diffⁿ betⁿ Go-Back-N ARQ and Selective Repeat ARQ.

<u>Go-Back-N ARQ</u>	<u>Selective Repeat ARQ</u>
(i) if it has a high error rate, it wastes a lot of bandwidth.	There is a loss of lot of bandwidth.
(ii) it is less complex.	it is more complex because it has to do sorting and searching as well.
(iii) it does not require sorting and searching.	iii) The Sort and Search operation is performed in it.
(iv) it is used more.	iv) it is used less because it is more complex.



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25/09